SC3102:
Signals, Systems and Transforms

Systems

Systems: Overview

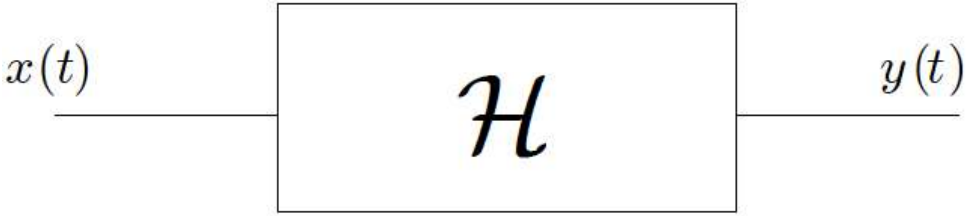
- **Systems**
  - Introduction to systems
  - Representations of systems
- **Classification of systems**
  - Continuous and discrete input systems
  - Systems with and without memory
  - Causal and noncausal systems
  - Linear and non-linear systems
  - Time variant and time-invariant systems
- **Linear time-invariant (LTI) systems**
  - Importance of LTI systems
  - General form of LTI systems
- **Analysis of LTI systems**
  - Discrete-time system analysis
  - Output of a discrete-time LTI systems
  - Impulse response and convolution
  - Continuous-time system analysis
  - Properties of LTI systems
- **Frequency response of LTI systems**
  - Discrete-time LTI systems
  - Continuous-time LTI systems
- **Modeling of LTI systems**
  - Examples
- **Summary**

Signals and systems

What is a system?

- Refers to any physical device that produces an output signal in response to an input signal

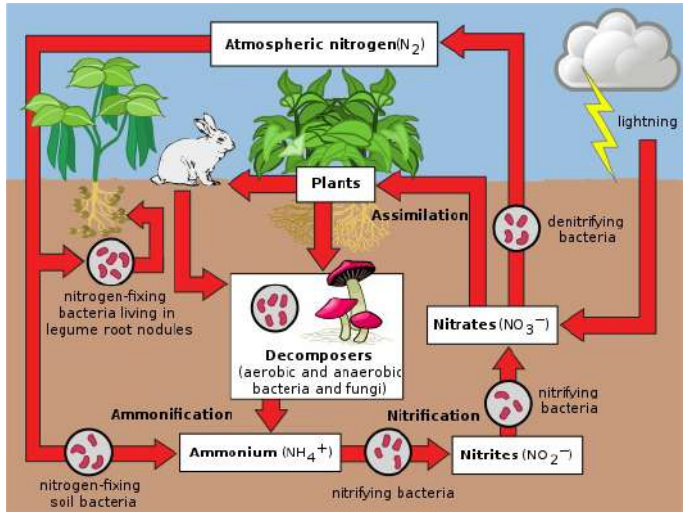
- Signals represents some variable of interest
- System is designed to process input signals and produce output signals, thus accomplishing a specific task

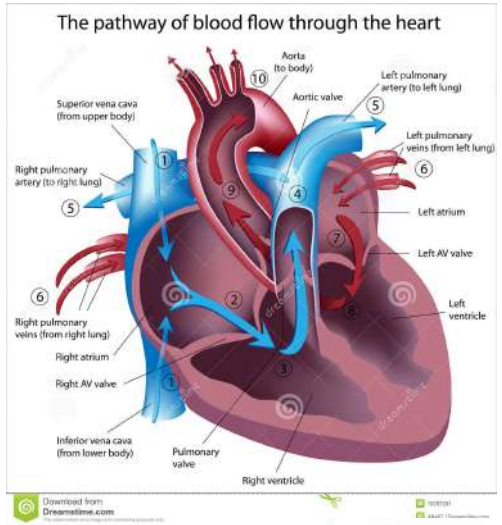


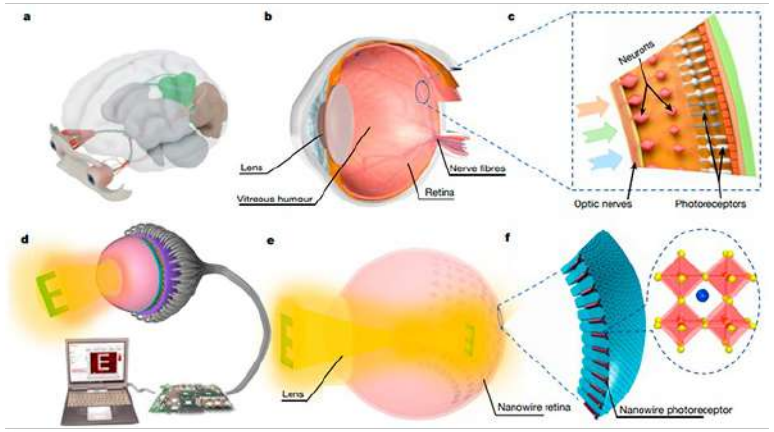
$$y(t) = \mathcal{H}[x(t)]$$

- We are surrounded by systems, both natural and artificial (man-made)
- Represented as functions, implemented by state machines, frequency responses, differential equations…..

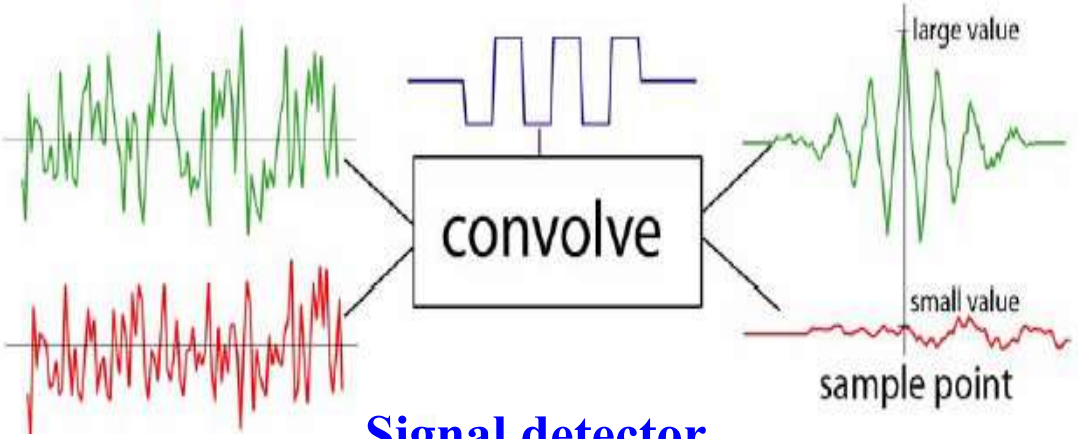
Systems: Examples for natural and artificial systems



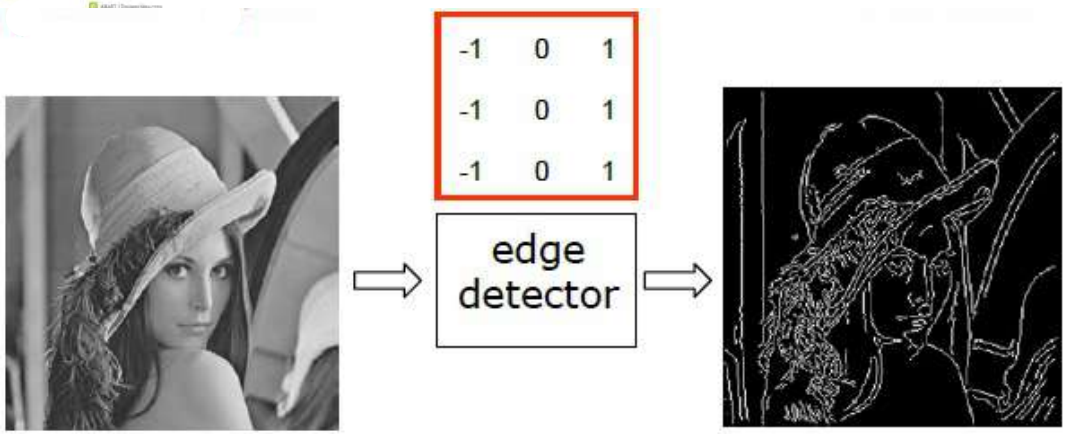




Human retina



Signal detector



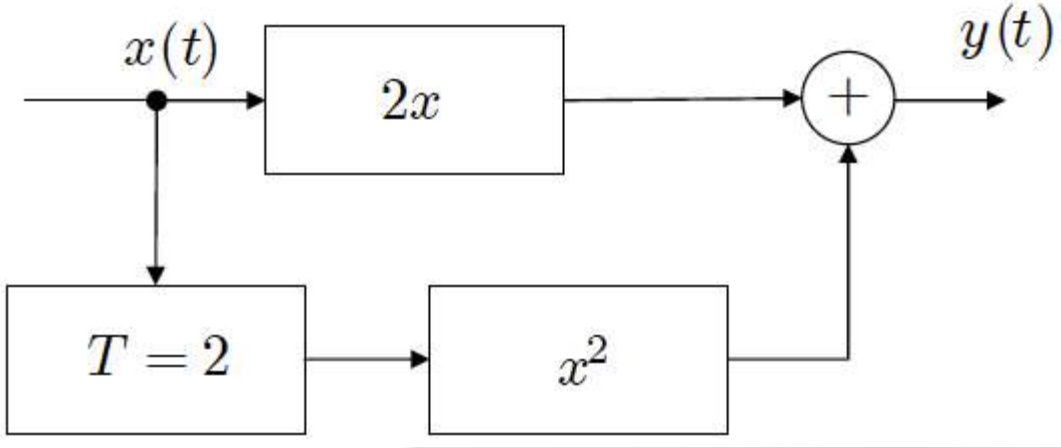
Edge detector

Representation of systems

- **Two main system representations**
  - (i) via input-output functional relationship
  - (ii) via system equation that involves current and past values of the output

- **Via input-output functional relationship $\boldsymbol{y = \mathcal{H}[x(t)]}$**
  - $\mathcal{H}$ is an arbitrary function the output depends only on the current, past and future input values

- **Example:**
  - $y(t) = 2x(t) + x^2(t-2)$

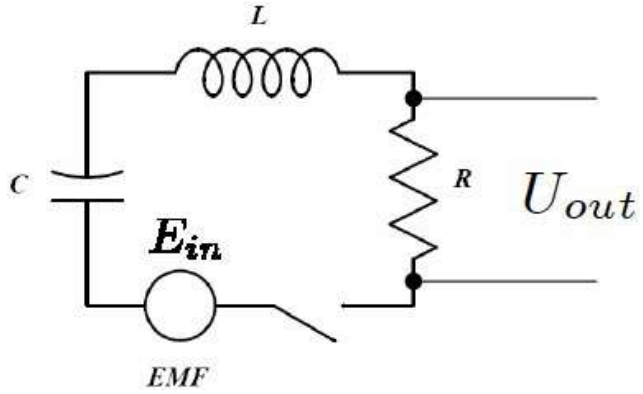


Representation of systems

- **Via system equation that involves current and past values of the output**

- **Examples**
  - Ex.1: Differential equation

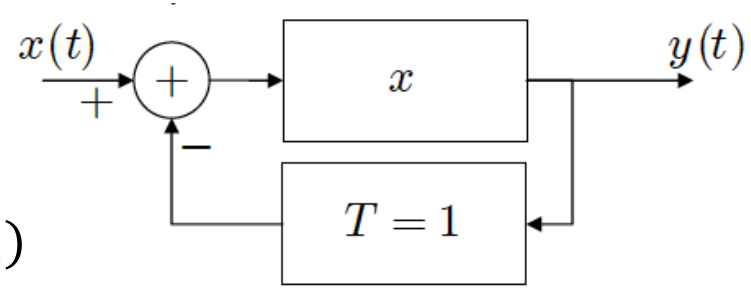
$$\frac{L}{R}U''_{out} + U'_{out} + \frac{1}{RC}U = E'_{in}$$



  - The order of the system is the order of the differential equation – highest derivative of the system output

  - Ex. 2: Feedback system

$$y(t) = x(t) - y(t-1)$$



Representation of systems

Conversion between system representations

- Representation via system equation can be converted to representation via input/output functional relationship.

Examples

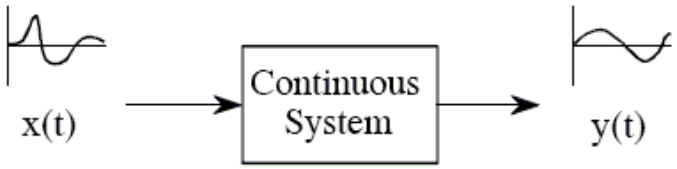
$$y'(t) = x(t) \quad \Longrightarrow \quad y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$$

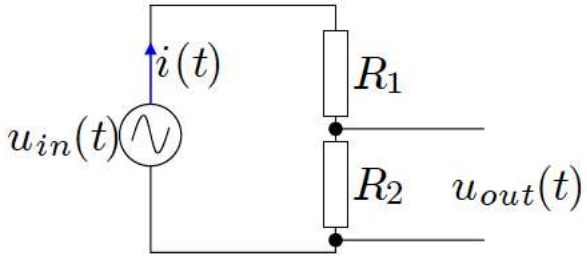
$$\begin{aligned} y(t) &= x(t) - 0.5y(t-1) \\ &= x(t) - 0.5\big(x(t-1) + 0.25y(t-2)\big) \\ &= x(t) - 0.5(x(t-1) + 0.25x(t-2) - 0.125y(t-3) + \cdots) \end{aligned}$$

- The reverse is generally not true

Classification of systems

Continuous and discrete time systems

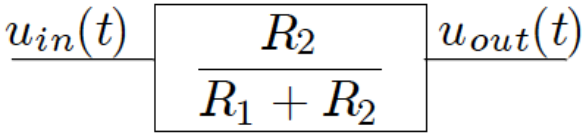




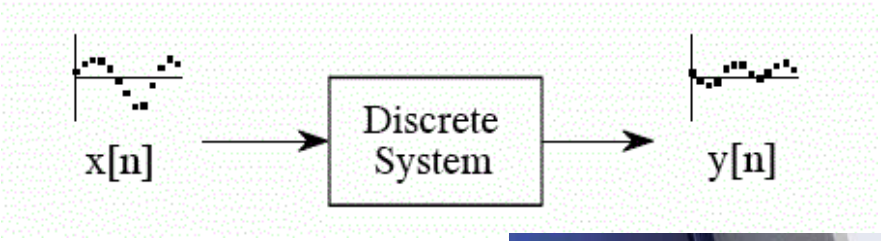
$$U_{in}(t) = i(t)(R_1 + R_2)$$

$$U_{out}(t) = i(t)R_2$$

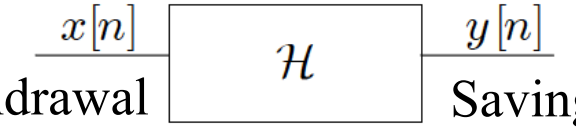
$$U_{out}(t) = \frac{R_2}{R_1+R_2} U_{in}(t)$$



Continuous time systems



Bank account



Deposit/withdrawal

Savings

If interest is 5%, then $y[n] = 1.05y[n-1] + x[n]$

Discrete time systems

Classification of systems

Memoryless systems and systems with memory

- Output of a **memoryless system** at time t_0 depends on its input at the same time t_0

$$U_{out}(t) = \frac{R_2}{R_1 + R_2} U_{in}(t)$$

- Output of a **system with memory** depends on its past and/or future inputs either
  - It can be explicit:

$$y = 2x(t) + \boxed{x^2(t-2)}$$ Past input

  - Or implicit:

$$y[n] = 1.05y[n-1] + x[n] \quad \Longrightarrow y[n-1] = 1.05y[n-2] + x[n-1]$$

$$y[n] = 1.1025y[n-2] + 1.05\boxed{x[n-1]} + x[n]$$

Past input

Classification of systems

Causal and noncausal systems

- Output of a **causal system** at time t_0 depends only on its input at times $t < t_0$

$$y[n] = 1.05y[n-1] + x[n]$$

- Output of a **noncausal system** at time t_0 can depends on its input at future times $t > t_0$

$$y[n] = 1.05y[n-1] + x[n+1]$$

- Noncausal systems are **physically unrealizable**
  - Only applies only to signals that are functions of time
  - Example: noncausal systems that process signals that are functions of coordinates are implementable (edge detector, for example)

Classification of systems

Linear and non-linear systems

- A system is called **linear** if adding or scaling inputs to the system results in adding or scaling the corresponding outputs

- For all constants a and input signals $x(t)$, $\mathcal{H}[ax(t)] = a\mathcal{H}[x(t)]$ **Homogeneity**

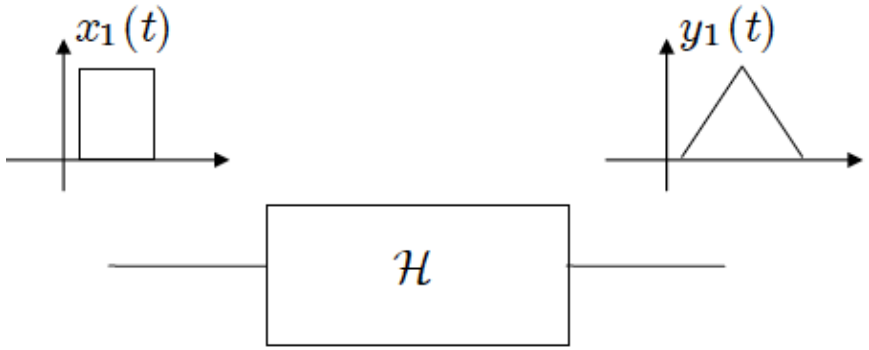
- For all input signals $x_1(t)$ and $x_2(t)$, $\mathcal{H}[x_1(t) + x_2(t)] = \mathcal{H}[x_1(t)] + \mathcal{H}[x_2(t)]$ **Additivity**

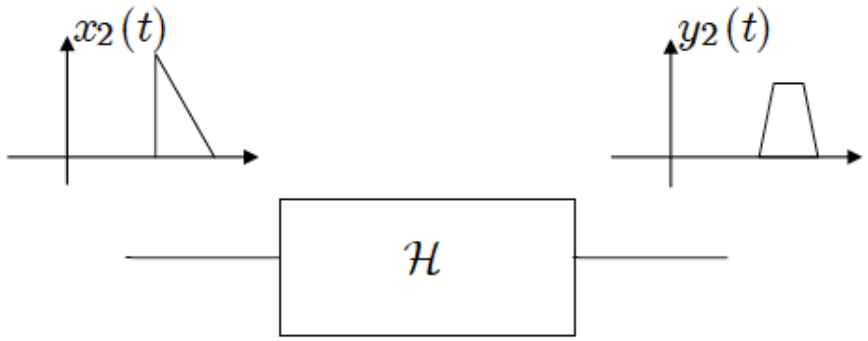
Principle of superposition

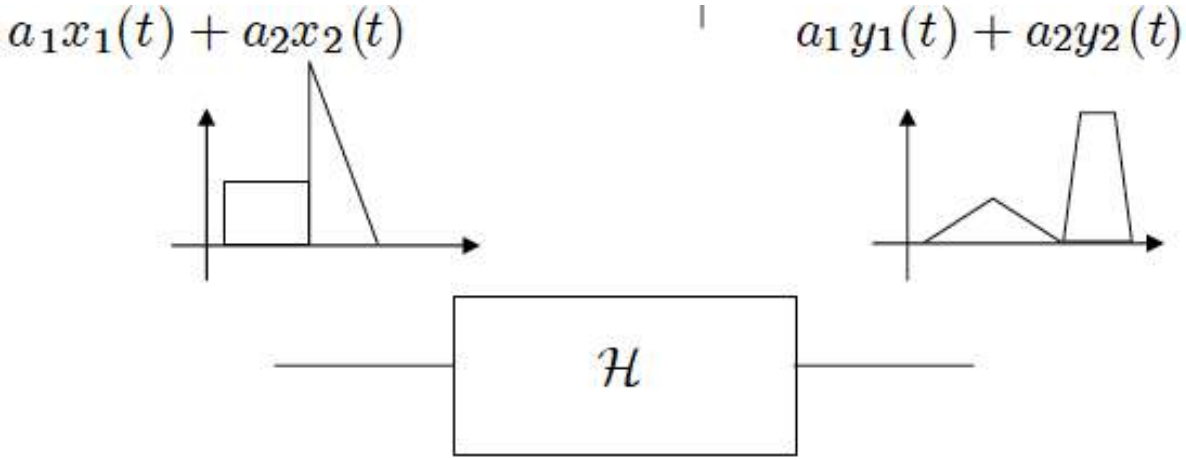
$$\boldsymbol{\mathcal{H}[a_1x_1(t) + a_2x_2(t)] = a_1\mathcal{H}[x_1(t)] + a_2\mathcal{H}[x_2(t)]}$$

- **Output of a linear system due to any input can easily be computed by representing the input signal via a linear combination of standard simple input signals with known output**

Linear systems: Example







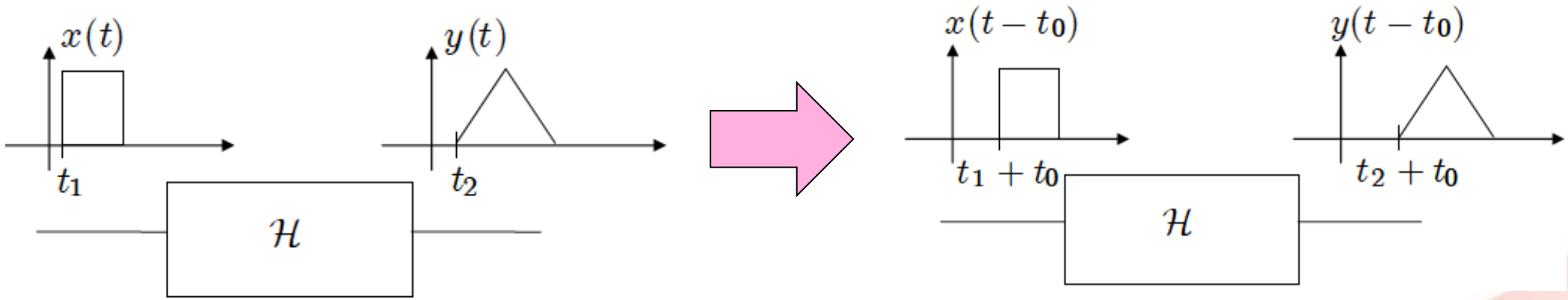
$$\mathcal{H}[a_1x_1(t) + a_2x_2(t)] = a_1\mathcal{H}[x_1(t)] + a_2\mathcal{H}[x_2(t)]$$

Classification of systems

- **Time variant and invariant systems**
  - A system is time-invariant if delaying the input by arbitrary t_0 produces the output delayed by t_0

- **Example**

$$y(t) = \mathcal{H}[x(t)] \quad \Rightarrow \quad \forall t_0, \qquad \mathcal{H}[x(t-t_0)] = y(t-t_0)$$



  - Time-invariance property allows quick computation of the output for delayed or advanced versions of the input

Linear and/or time-invariant systems: Example

- **Determine whether the system described by the following differential equation is linear and/or time-invariant:**

$$y' + ty = x(t)$$

- **Linear:**
  - Let $y_1(t)$ and $y_2(t)$ be the responses to $x_1(t)$ and $x_2(t)$.
  - $y_1' + ty_1 = x_1(t)$ and $y_2' + ty_2 = x_2(t)$
  - Multiplying these equations with a_1 and a_2 respectively and then add them up gives
  - $a_1 \frac{d}{dt} y_1(t) + a_1 t y_1 + a_2 \frac{d}{dt} y_2(t) + a_2 t y_2 = a_1 x_1(t) + a_2 x_2(t)$
  - $\frac{d}{dt}\big(a_1 y_1(t) + a_2 y_2(t)\big) + t\big(a_1 y_1(t) + a_2 y_2(t)\big) = \big(a_1 x_1(t) + a_2 x_2(t)\big)$
  - Let $y_0(t)$ be the response to $a_1 x_1(t) + a_2 x_2(t)$, then $y_0' + t y_0 = a_1 x_1(t) + a_2 x_2(t)$

Linear and/or time-invariant systems: Example

$y' + ty = x(t)$

- Clearly, $y_0(t)$ and $a_1 y_1(t) + a_2 y_2(t)$ satisfy the same differential equation,
- Hence the system is **linear**.

Time-variant

- Let $y(t)$ be the response to $x(t)$, then $y' + ty = x(t)$
- Let $y_0(t)$ be the response to $x(t - t_0)$, then $y_0' + ty_0 = x(t - t_0)$
- On the other hand, delayed output $y(t - t_0)$ satisfies $y'(t - t_0) + (t - t_0)y(t - t_0) = x(t - t_0)$
- $y_0(t)$ and $y(t - t_0)$ are solutions of different differential equations and hence $y_0(t) \neq y(t - t_0)$
- So the system is **Time-variant**

- **<u>Note:</u>** The system defined by an ordinary differential equation (ODE) is time invariant if all coefficients of the ODE are constant. Otherwise, the system is time-variant.

Linear and/or time-invariant systems: Example

- **Determine whether the system described by the following system equation is linear and/or time-invariant:**

$$y(t) = y(t-1) + 5 + x(t)$$

- **Linear:**
  - Let $y_1(t)$ and $y_2(t)$ be the responses to $x_1(t)$ and $x_2(t)$.
  - $y_1(t) = y_1(t-1) + 5 + x_1(t)$ and $y_2(t) = y_2(t-1) + 5 + x_2(t)$
  - Multiplying these equations with a_1 and a_2 respectively and then add them up gives
  - $a_1y_1(t) + a_2y_2(t) = a_1y_1(t-1) + a_2y_2(t-1) + (5a_1 + 5a_2) + a_1x_1(t) + a_2x_2(t)$

  - Let $y_0(t)$ be the response to $a_1x_1(t) + a_2x_2(t)$, then $y_0(t) = y_0(t-1) + 5 + a_1x_1(t) + a_2x_2(t)$
  - Clearly, $y_0(t)$ and $a_1y_1(t) + a_2y_2(t)$ are solutions of different equations since $5a_1 + 5a_2 \neq 5$

  - Hence the system is **nonlinear**

Linear and/or time-invariant systems: Example

$$y(t) = y(t-1) + 5 + x(t)$$

- **Time-invariant:**
  - Let $y(t)$ be the response to $x(t)$ and hence $y(t) = y(t-1) + 5 + x(t)$
  - Delayed output $y(t-t_0)$ is obtained through the change of variable $t \to t - t_0$
  - That is, $y(t-t_0) = y(t-t_0-1) + 5 + x(t-t_0)$
  - Let $y_0(t)$ be the response to $x(t-t_0)$, then $y_0(t) = y_0(t-1) + 5 + x(t-t_0)$
  - Clearly, $y_0(t)$ and $y(t-t_0)$ satisfy the same equation
  - Hence the system is **time-invariant**

Linear and/or time-invariant systems: Example

- **Let $g(t)$ be an arbitrary function of time. Determine whether the system described by $y(t) = g(t)x(t)$ is linear and/or time-invariant.**

- **Answer:**
  - Let $y_1(t)$ and $y_2(t)$ be the responses to $x_1(t)$ and $x_2(t)$.
  - $y_1(t) = g(t)x_1(t)$ and $y_2(t) = g(t)x_2(t)$
  - Multiplying these equations with a_1 and a_2 respectively and then add them up gives
  - $a_1y_1(t) + a_2y_2(t) = a_1g(t)x_1(t) + a_2g(t)x_2(t) = g(t)\big(a_1x_1(t) + a_2x_2(t)\big)$
  - Let $y_0(t)$ be the response to $a_1x_1(t) + a_2x_2(t)$, then $y_0(t) = g(t)\big(a_1x_1(t) + a_2x_2(t)\big)$
  - Hence the system is **linear**
  - The response to $x(t - t_0)$ is $g(t)x(t - t_0) \neq y(t - t_0) = g(t - t_0)x(t - t_0)$
  - Hence the system is **time-variant**

Linear and/or time-invariant systems: Example

- **Determine whether the system described by $y(t) = x(t) + K$ is linear and/or time-invariant.**

- **Answer:**
  - Let $y_1(t)$ and $y_2(t)$ be the responses to $x_1(t)$ and $x_2(t)$.
  - $y_1(t) = x_1(t) + K$ and $y_2(t) = x_2(t) + K$
  - Let $y_0(t)$ be the response to $a_1x_1(t) + a_2x_2(t)$, then $y_0(t) = a_1x_1(t) + a_2x_2(t) + K$
  - Note that $y_0(t) \neq a_1y_1(t) + a_2y_2(t) = \big(a_1x_1(t) + a_2x_2(t)\big) + (a_1 + a_2)K$
  - Hence the system is **nonlinear**
  - The response to $x(t - t_0)$ is $x(t - t_0) + K = y(t - t_0)$
  - Hence the system is **time-invariant**

Classification of systems

- **Linear time-invariant systems**
  - System is called linear time-invariant (LTI) if it is both linear and time-invariant

- **Why special attention to LTI systems?**
  - Majority of naturally occurring and man-made systems can be modeled as LTI systems
  - Example: Signal detector, image edge detector, human brain, etc..
  - Allows simple system analysis and system design

  - **System analysis**: understand how a given system operates (Ex: analyzing a photograph)
  - Complete understanding leads to the prediction of output signal from any given input signal

  - **System design**: reverse of the signal analysis (Ex: design a system that determines image edges)
  - Given desired system behavior, one should be able to design an accurate system

Linear time-invariant systems

General form of a LTI system

- Output of a LTI system is a linear combination of delayed and/or advanced version of the input
- Any LTI system can be described by the input/output relationship of the form

$$y(t) = C_1 x(t - t_1) + C_2 x(t - t_2) + \cdots + C_N x(t - t_N) \Rightarrow y(t) = \sum_{k=1}^{N} C_k x(t - t_k)$$

where C_k is a constant, t_k can be positive or negative and N can be finite or infinite

- Can represent an input to LTI system in terms of linear combinations of a set of basic signals
- Use superposition to compute the system output in terms of its responses to these basic signals
- **With the properties of superposition and time-invariance, can develop a complete characterization of any LTI systems in terms of its response to a unit impulse**

Linear time-invariant systems

- **Show that the given system $y(t) = \sum_{k=1}^{N} C_k x(t - t_k)$ represents a linear time-invariant system**
  - Note: C_k is a constant, t_k can be positive or negative and N can be finite or infinite

- **Answer:**
  - Let $y_1(t)$ and $y_2(t)$ be the responses to $x_1(t)$ and $x_2(t)$.
  - $y_1(t) = \sum_{k=1}^{N} C_k x_1(t - t_k)$ and $y_2(t) = \sum_{k=1}^{N} C_k x_2(t - t_k)$
  - Let $y_0(t)$ be the response to $a_1 x_1(t) + a_2 x_2(t)$, then
    $y_0(t) = \sum_{k=1}^{N} C_k \left(a_1 x_1(t - t_k) + a_2 x_2(t - t_k)\right) = a_1\, y_1(t) + a_2\, y_2(t)$
  - Hence the system is **linear**.

  - The response to $x(t - t_0)$ is $\sum_{k=1}^{N} C_k x(t - t_k - t_0) = y(t - t_0)$
  - Hence the system is **time-invariant**.

Analysis of Linear time-invariant systems

Discrete-time system analysis

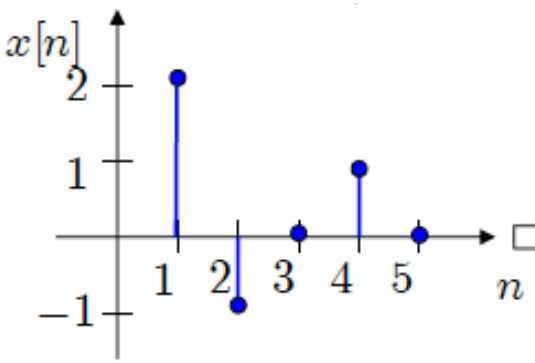
- Assume we have a discrete-time LTI system.
- How to represent the output of the system in the general form?

- Goal: find the output of the system for any arbitrary input?

Discrete-time system analysis: Example

Discrete-time system analysis

- Goal: find the output of the system for any arbitrary discrete input signal?
- From the property of Kronecker delta function, $x[n] = \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k]$



$$x[n] = x[1]\delta[n-1] + x[2]\delta[n-2] + x[4]\delta[n-4]$$

$$\boxed{x[n] = 2\delta[n-1] - \delta[n-2] + \delta[n-4]}$$ **Input signal**

- **Any discrete-time signal can be represented as a linear combination of Kronecker delta functions**
- Apply Kronecker delta function $\delta[n]$ as input of the given system and obtain the output $h[n]$

Discrete-time system analysis: Example

- **Discrete-time system analysis**
  - Apply Kronecker delta function $\delta[n]$ as input of the given system and obtain the output $h[n]$

$h[n] = \mathcal{H}(\delta[n])$

$\boldsymbol{y[n] = \mathcal{H}(x[n])}$

Discrete-time system analysis: Example

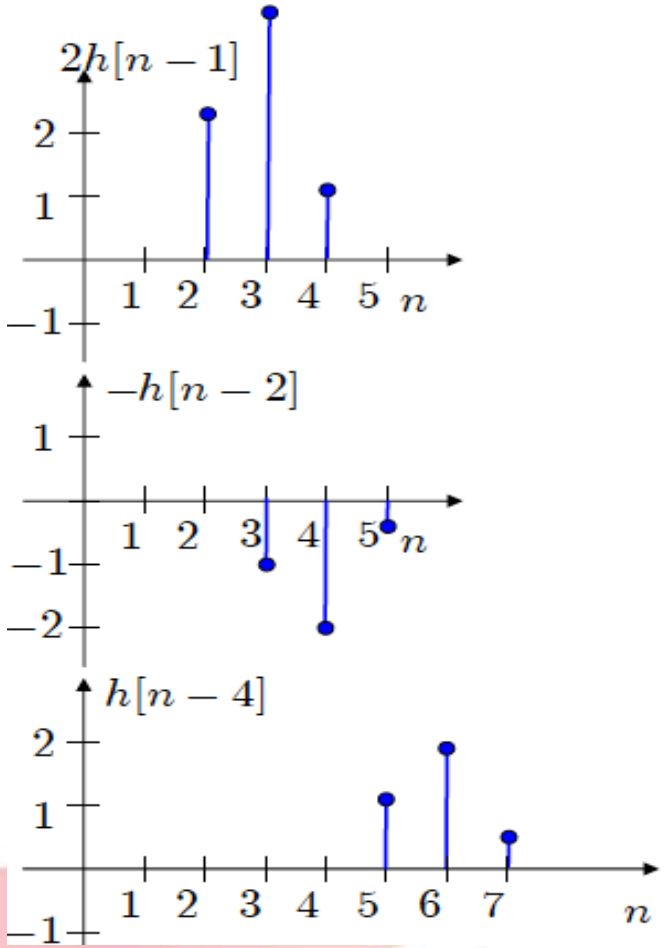
Discrete-time system analysis

- We have, $x[n] = 2\delta[n-1] - \delta[n-2] + \delta[n-4]$
- $y[n] = \mathcal{H}(x[n])$

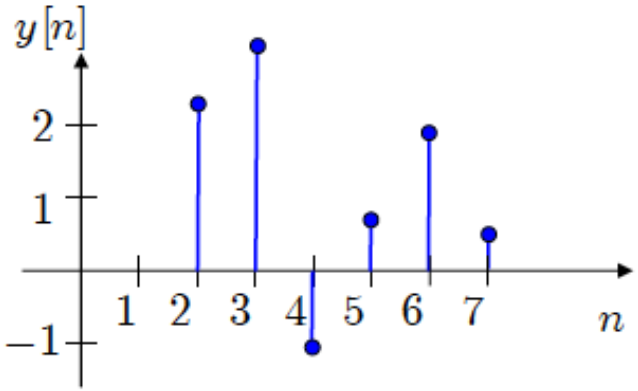
$$= \mathcal{H}(2\delta[n-1] - \delta[n-2] + \delta[n-4])$$

$$= \mathcal{H}(2\delta[n-1]) - \mathcal{H}(\delta[n-2]) + \mathcal{H}(\delta[n-4])$$

$$= 2h[n-1] - h[n-2] + \delta[n-4]$$



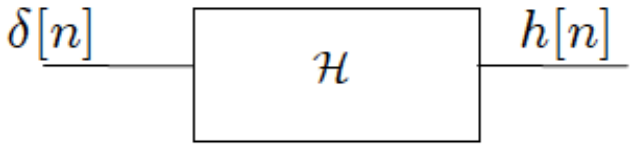
$$\boxed{h[n] = \mathcal{H}(\delta[n])}$$



Analysis of Linear time-invariant systems

Computing the output of a discrete-time LTI system

- Let $h[n]$ be the response of the system to $\delta[n]$, then $h[n] = \mathcal{H}(\delta[n])$
- **Response to an arbitrary signal $x[n]$:**



$$y[n] = \mathcal{H}(x[n]) = \mathcal{H}\left(\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\right)$$

$$= \sum_{k=-\infty}^{\infty} \mathcal{H}(x[k]\delta[n-k]) \qquad \textbf{Linearity}$$

$$= \sum_{k=-\infty}^{\infty} x[k]\mathcal{H}(\delta[n-k]) \qquad \textbf{Linearity}$$

$$= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \qquad \textbf{Time-invariance}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Analysis of Linear time-invariant systems

Output of a discrete-time LTI system

- Output of a LTI system when input is $x[n]$ is $y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$

- Change of variable $n - k \rightarrow m$, $k \rightarrow n - m$

$$y[n] = \sum_{m=-\infty}^{\infty} x[n-m]h[m] = \cdots + h[1]\mathrm{x}[n-1] + h[2]\mathrm{x}[n-2] + \cdots$$

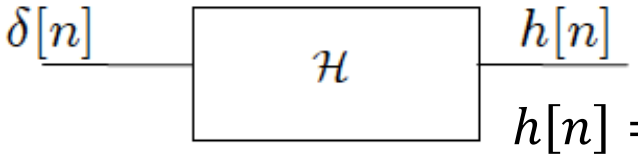
$$\boxed{\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\ &= \sum_{m=-\infty}^{\infty} x[n-m]h[m] \end{aligned}}$$

- It is identical to the general form of a linear time-invariant system
  $y(t) = C_1 x(t - t_1) + C_2 x(t - t_2) + \cdots + C_N x(t - t_N)$
  if we take $C_1 = h[i]$

Impulse response and convolution for a discrete-time system

Impulse response of a system

- Impulse response $h[n]$ of a discrete-time LTI system is the response of the system to the Kronecker delta function



$$h[n] = \mathcal{H}(\delta[n])$$

- Response is taken at zero initial conditions

Convolution

- Given the input signal $x[n]$, the output $y[n]$ of a discrete-time LTI system can be computed by **convolving** $x[n]$ with $h[n]$:

$$y[n] = h[n] * x[n] = x[n] * h[n] = \boxed{\sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} h[k]x[n-k]}$$

Convolution is commutative

Analysis of Linear time-invariant systems

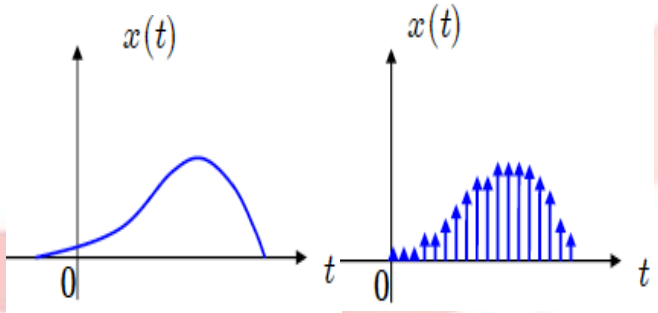
Continuous-time system analysis

- Goal: Find the output of the system for any arbitrary input

- Put the delta function $\delta(t)$ on the input and measure the output $h(t)$

- We have, $x(t) = \int_{-\infty}^{z=\infty} x(\tau)\,\delta(t-\tau)d\tau$

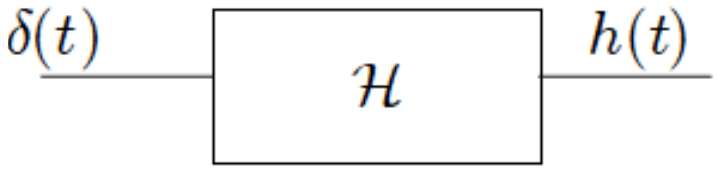
- **Any continuous-time signal can be represented as a finite weighted sum of shifted delta functions**



Analysis of Linear time-invariant systems

Computing the output of continuous-time LTI system

- Let $h(t)$ be the response of the system to $\delta(t)$, then $h(t) = \mathcal{H}\big(\delta(t)\big)$
- **Response to an arbitrary signal $x(t)$:**



$$y(t) = \mathcal{H}\big(x(t)\big) = \mathcal{H}\left(\int_{-\infty}^{+\infty} x(\tau)\delta(t-\tau)d\tau\right)$$

$$= \int_{-\infty}^{+\infty} \mathcal{H}\big(x(\tau)\delta(t-\tau)\big)d\tau \qquad \textbf{Linearity}$$

$$= \int_{-\infty}^{+\infty} x(\tau)\mathcal{H}\big(\delta(t-\tau)\big)d\tau \qquad \textbf{Linearity}$$

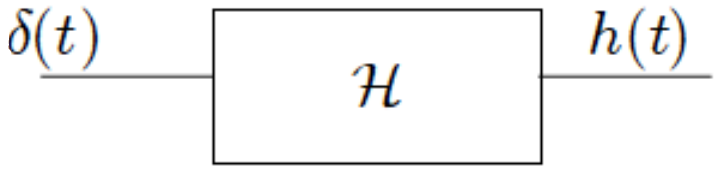
$$= \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau \qquad \textbf{Time-invariance}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Impulse response and convolution for Continuous-time systems

Impulse response of a system

- Impulse response of a continuous-time LTI system is the response of the system to delta function



$$h(t) = \mathcal{H}\big(\delta(t)\big)$$

Convolution

- Given the input $x(t)$, the output of a continuous-time LTI system $y(t)$ can be computed by **convolving** $x(t)$ with $h(t)$

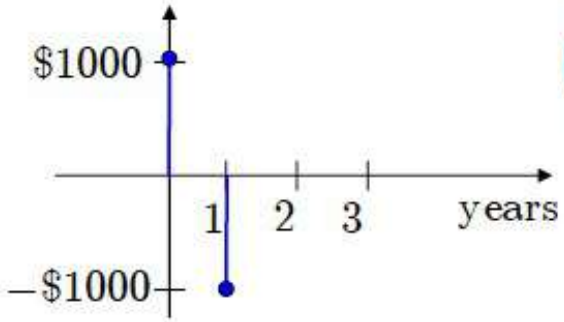
$$y(t) = h(t) * x(t) = x(t) * h(t) \boxed{= \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau}$$

Convolution is commutative

Output of LTI systems: Example

- **A bank offers a saving account with 5% interest rate. A customer makes a deposit of $1000 in the first year and a withdrawal of $1000 in the second year. Determine the account value as a function of time.**

- **Answer:**
  - Let us approach the problem from a signals and system perspective
  - The account balance in a saving account with 5% interest is $y[n] = 1.05y[n-1] + x[n]$



$$y[n] = 1.05y[n-1] + x[n]$$

Output of LTI systems: Example

$$y[n] = 1.05y[n-1] + x[n]$$

- Let $x[n] = \delta[n]$. Then, $h[n] = 1.05h[n-1] + \delta[n]$

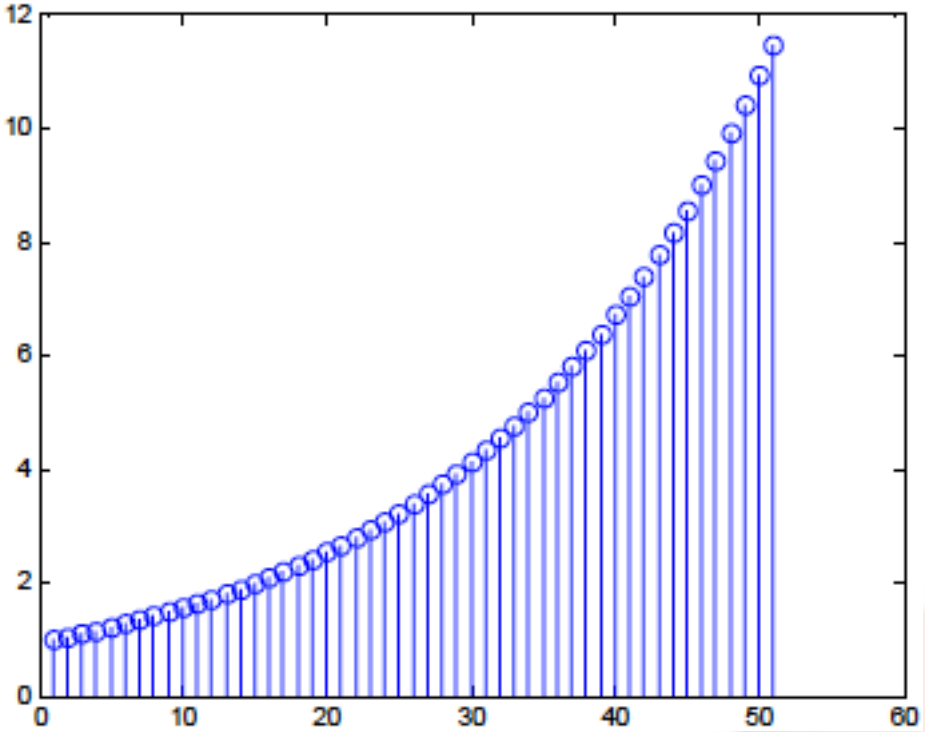
  $h[n] = 0, \quad \mathrm{n} < 0$ **causality**

  $h[0] = 1.05h[-1] + 1 = 1$

  $h[1] = 1.05h[0] + 0 = 1.05$

  $h[2] = 1.05h[1] = 1.05 * 1.05 = 1.05^2$

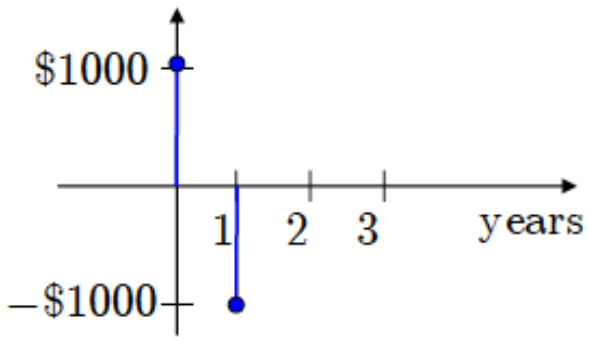
  $h[n] = 1.05^n$



- Impulse response of a bank account system is a **discrete-time exponentially increasing function**

Output of LTI systems: Example

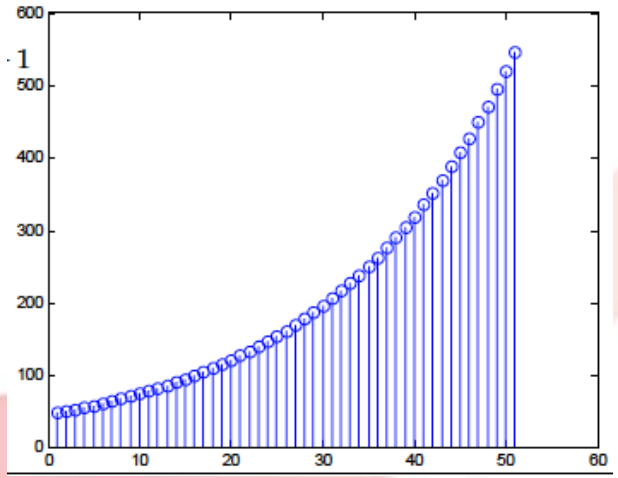
- **Computing the output**
  - Input: $x[n] = 1000\delta[n] - 1000\delta[n-1]$
  - $h[n] = 1.05h[n-1] + \delta[n]$
  - The output $y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$



$$y[n] = 1000 \sum_{k=-\infty}^{+\infty} (\delta[n] - \delta[n-1])h[n-k] = 1000h[n] - 1000h[n-1]$$

$$y[n] = 1000(h[n] - h[n-1])$$

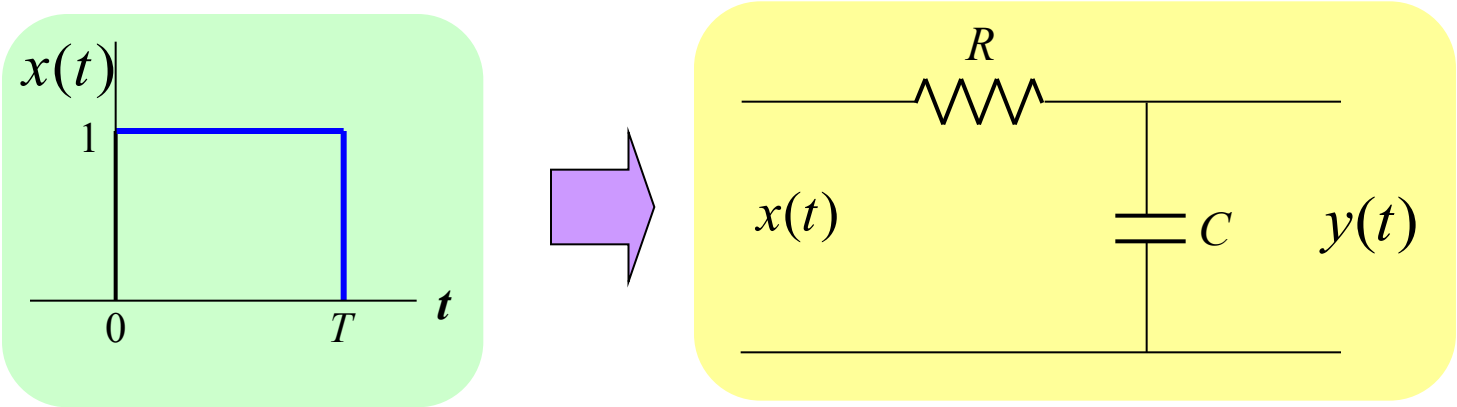
$$y[n] = 1000(1.05^n - 1.05^{n-1}) = 50(1.05^{n-1})$$



- In 50 years, the bank account will have more than $500 !

Output of LTI systems: Example

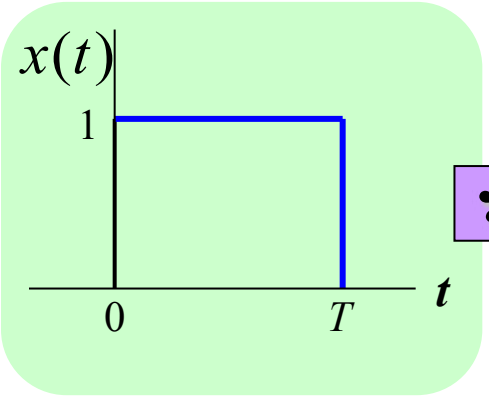
- **Consider the RC network in the following figure. Find the output $y(t)$ if the input $x(t)$ is a rectangular pulse as shown below, and $RC = 1$**



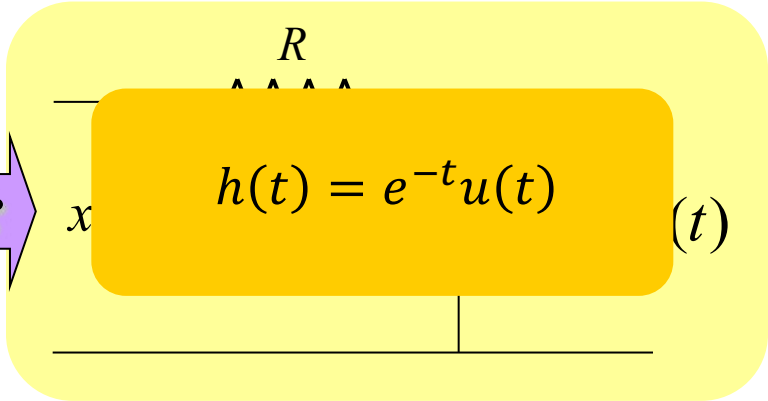
- Note: The system function $h(t)$ for the given RC network is $h(t) = e^{-t}u(t)$ when RC = 1

Output of LTI systems: Example

- **Given:**
  - Input $\boldsymbol{x(t)}$ is a rectangular pulse and system response for the RC network is $h(t) = e^{-t}u(t)$

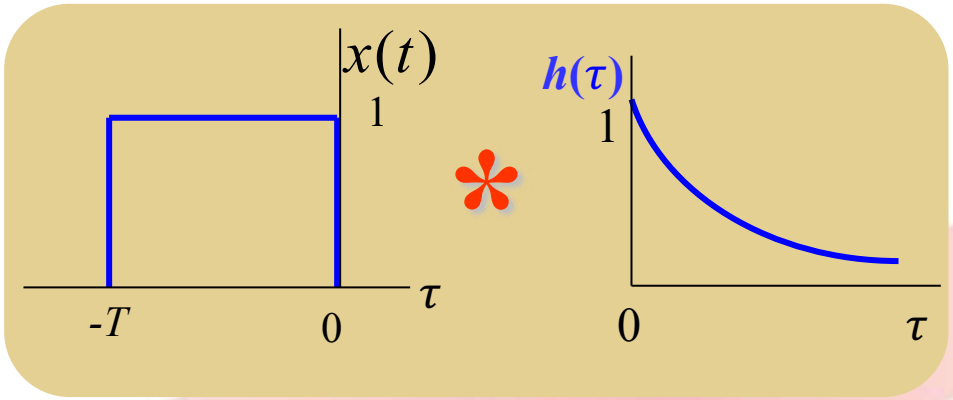


$*$



$$y(t) = x(t) * h(t)$$
$$= h(t) * x(t)$$

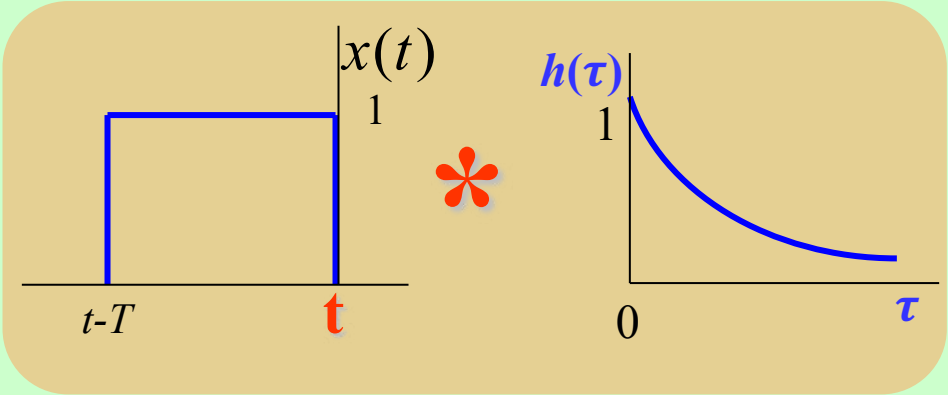
$$y(t) = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau$$
$$= \int_{0}^{+\infty} e^{-\tau}x(t-\tau)d\tau$$



Output of LTI systems: Example

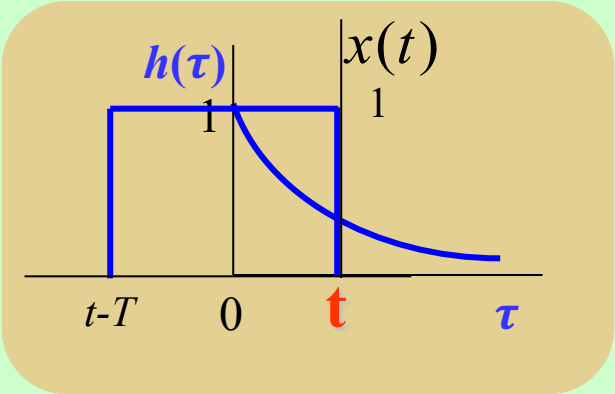
For $t < 0$,

$$y(t) = 0$$



For $0 \le t \le T$,

$$y(t) = \int_0^t e^{-\tau} d\tau$$
$$= -e^{-\tau}\Big|_0^t$$
$$= -(e^{-t} - e^0)$$
$$= 1 - e^{-t}$$



Output of LTI systems: Example

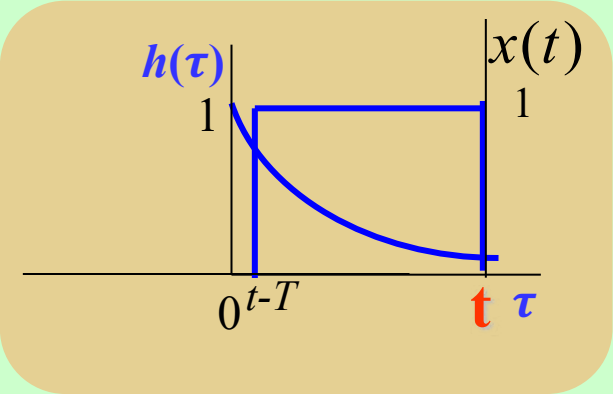
For $t > T$,

$$y(t) = \int_{t-T}^{t} e^{-\tau} d\tau = [e^{T} - 1]e^{-t}$$

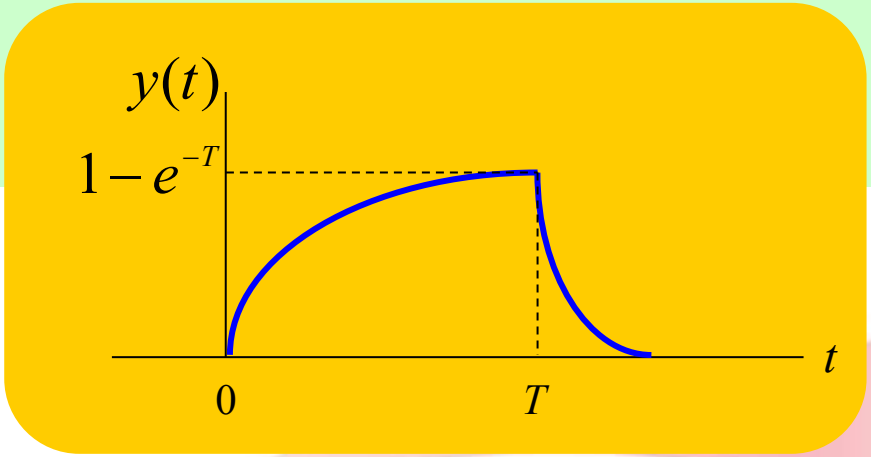
$$= -e^{-\tau}\Big|_{t-T}^{t}$$

$$= -\left(e^{-t} - e^{-(t-T)}\right)$$

$$= e^{-t}(e^{T} - 1)$$



Putting all together,

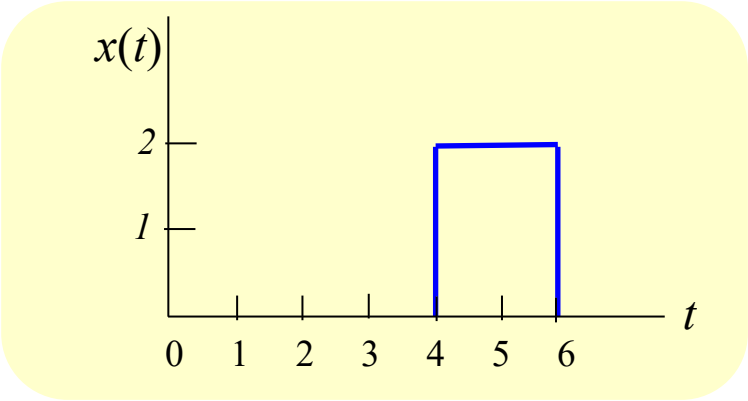


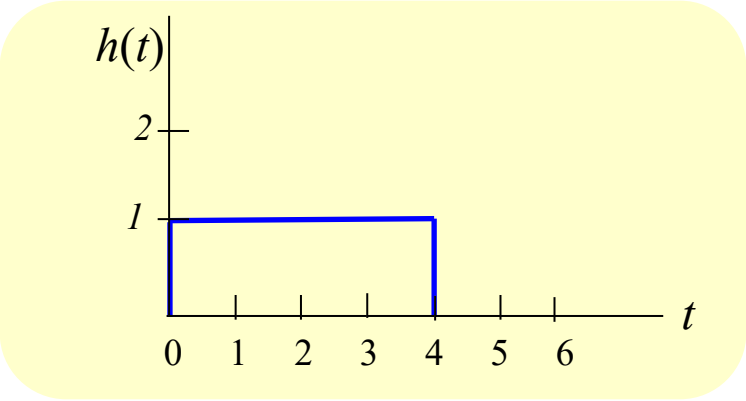
Output of LTI systems: Example

- **Find the convolution of the following signals**

$$x(t) = 2\Pi\left(\frac{t-5}{2}\right), \qquad h(t) = \Pi\left(\frac{t-2}{4}\right)$$

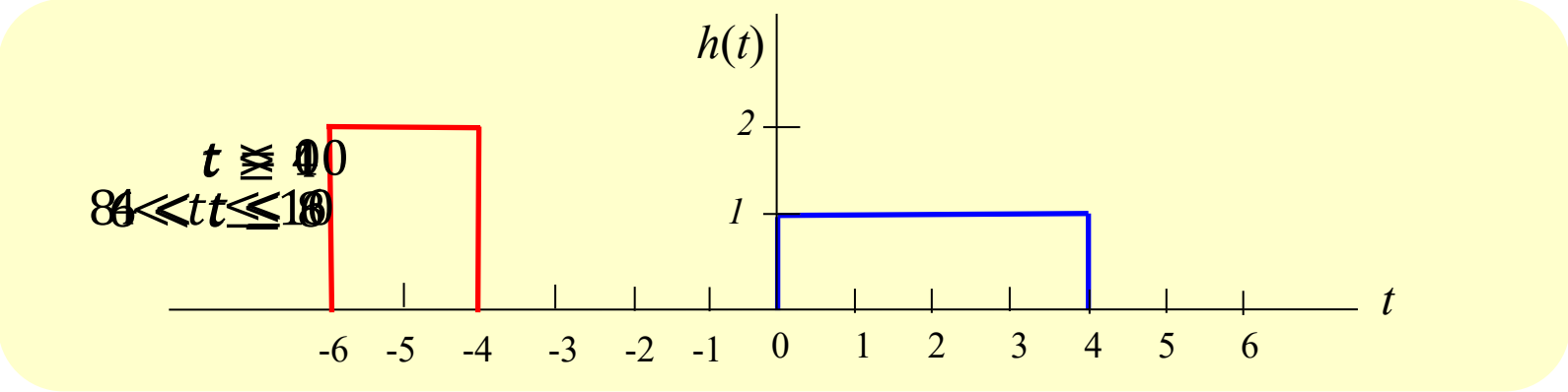
- **Answer:**

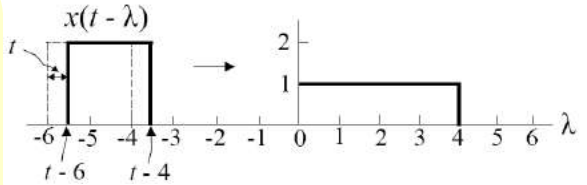


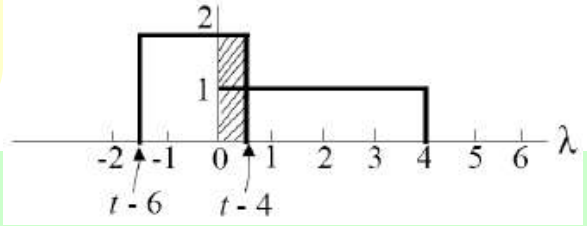


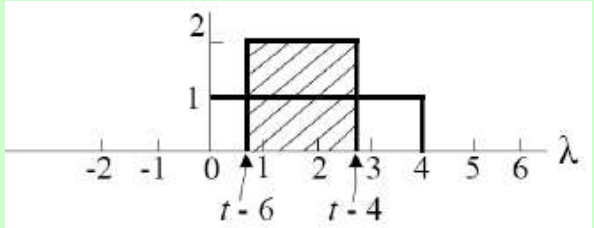
$$x(t) * h(t) = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau$$

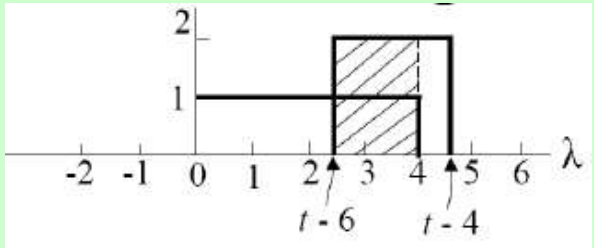
Output of LTI systems: Example











Case 1: $t \leq 4 \quad \Longrightarrow \quad y(t) = 0$

Case 2: $4 < t \leq 6 \;\Longrightarrow y(t) = 2(t-4)$

Case 3: $6 < t \leq 8 \;\Longrightarrow$

$$y(t) = 2[(t-4)-(t-6)] = 4$$

Case 4: $8 < t \leq 10 \;\Longrightarrow$

$$y(t) = 2[4-(t-6)] = 2(10-t)$$

Case 5: $t > 10 \quad \Longrightarrow \quad y(t) = 0$

Output of LTI systems: Example

Case 1: $t \leq 4 \quad \Longrightarrow \quad y(t) = 0$

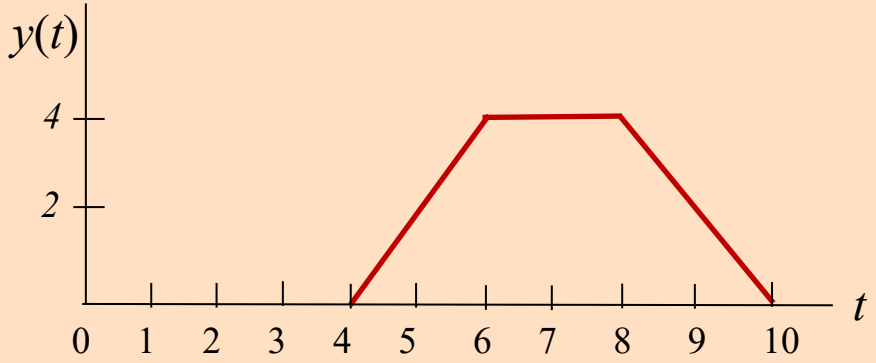
Case 2: $4 < t \leq 6 \;\Longrightarrow y(t) = 2(t-4)$

Case 3: $6 < t \leq 8 \;\Longrightarrow y(t) = 2[(t-4)-(t-6)] = 4$

Case 4: $8 < t \leq 10 \;\Longrightarrow y(t) = 2[4-(t-6)] = 2(10-t)$

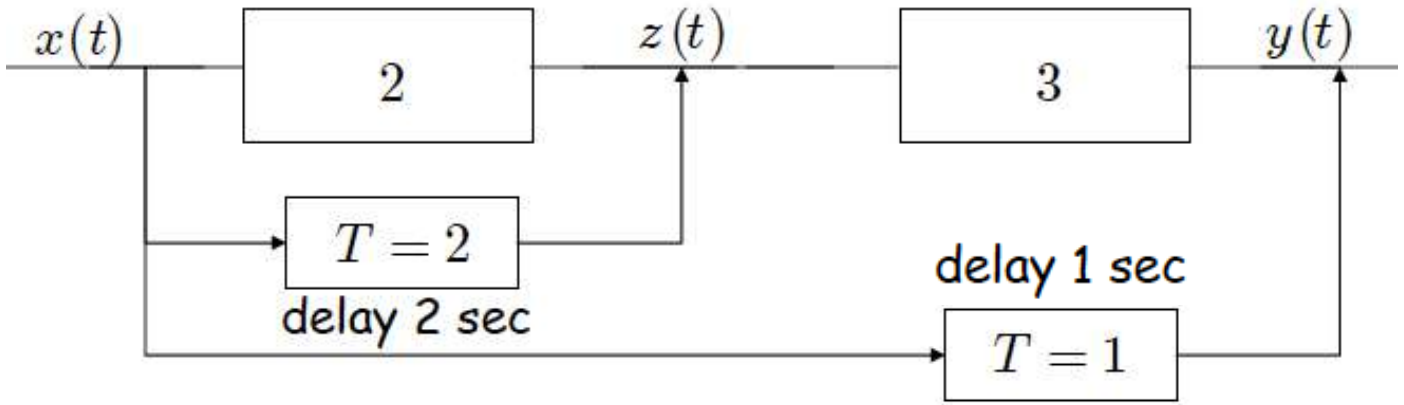
Case 5: $t > 10 \quad \Longrightarrow \quad y(t) = 0$

$$y(t) = \begin{cases} 0, & t \leq 4 \\ 2(t-4), & 4 < t \leq 6 \\ 4, & 6 < t \leq 8 \\ 2(10-t), & 8 < t \leq 10 \\ 0, & t > 10 \end{cases}$$



Output of LTI systems: Example

- **Combination of systems**
  - Determine the impulse response of the following system:



- **Answer:**

$$z(t) = 2x(t) + x(t-2)$$

$$y(t) = 3z(t) + x(t-1) = 6x(t) + x(t-1) + 3x(t-2)$$

  - Impulse response: $\mathrm{h}(t) = 6\delta(t) + \delta(t-1) + 3\delta(t-2)$

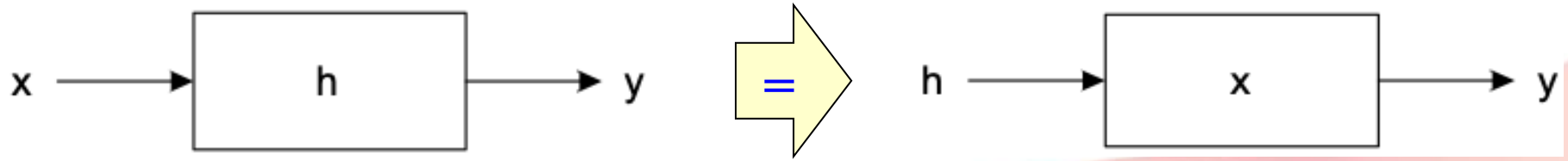
Properties of LTI systems

The commutative property of LTI systems

- Convolution in both continuous and discrete time is a commutative operation.
- That is,

$$x[n] * h[n] = h[n] * x[n] = \sum_{k=-\infty}^{+\infty} x[k]\, h[n-k] = \sum_{k=-\infty}^{+\infty} h[k]\, x[n-k]$$

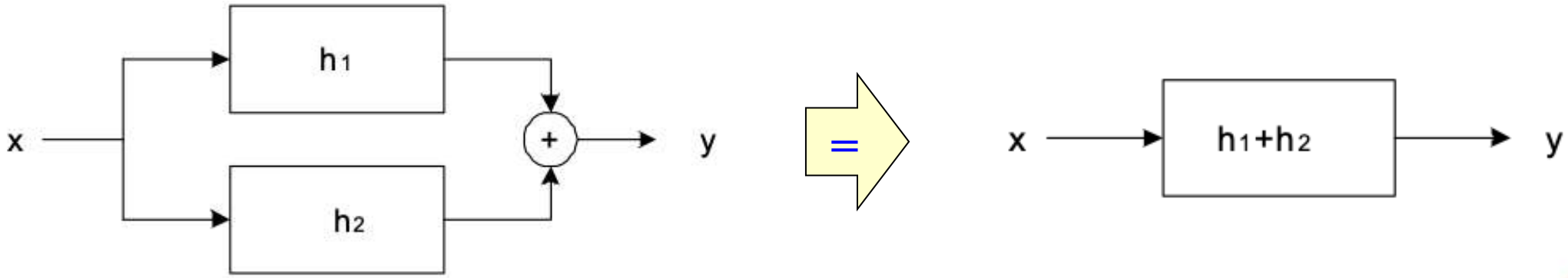
$$x(t) * h(t) = h(t) * x(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau$$



Properties of LTI systems

Distributive property of LTI systems

- Summing the outputs of two systems is equivalent to a system with an impulse response equal to the sum of the impulse response of the two individual systems
- That is, $x * (h_1 + h_2) = x * h_1 + x * h_2$

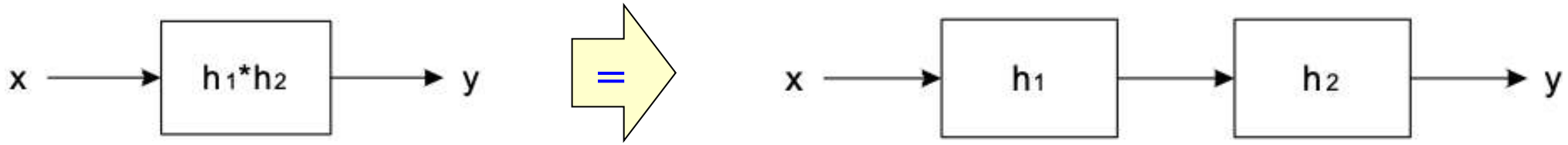


- Distributive property of convolution can be exploited to break a complicated convolution into several simpler ones.

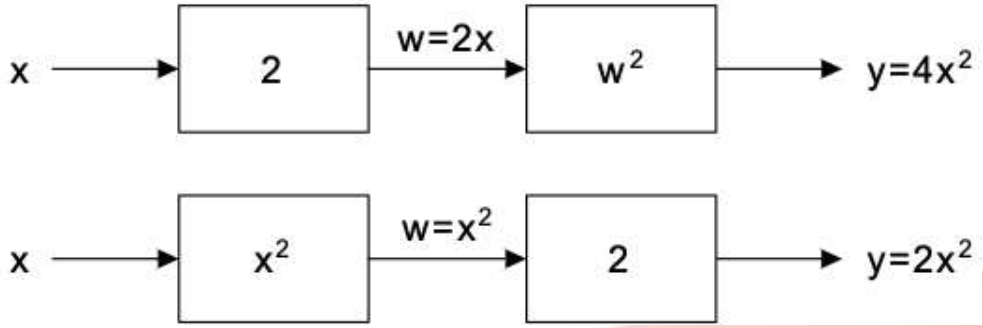
Properties of LTI systems

Associative property

- For LTI systems, the change of order of the cascaded systems will not affect the response.
- That is, $x * (h_1 * h_2) = (x_1 * h_1) * h_2$



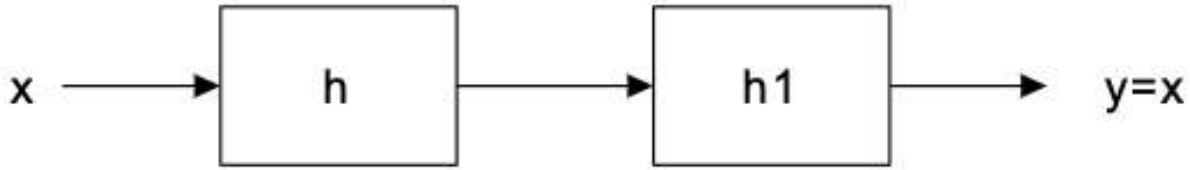
- Note: for nonlinear systems, the order of cascaded systems in general cannot be changed.



Properties of LTI systems

Invertibility of LTI systems

- A system S is invertible if and only if there exists an inverse system S^{-1}such that $S^{-1}S$ is an identity system



- Here, the overall impulse response is h_1 must satisfy for it to be the impulse response of the inverse system, namely $h * h_1 = \delta$

- Application:
  - Channel equalisation in communication system: The transmitted communication signal will be processed at the receiver through a filter whose impulse response is designed to be inverse of the impulse response of the communication channel

Properties of LTI systems

Stability of LTI systems

- A system is stable if every bounded input produces a bounded output

- For a discrete-time LTI system, the input $x[n]$ is bounded in magnitude. i.e., $|x[n]| \leq B$, for all n
- The input $x[n]$ is applied to an LTI system with unit impulse response $h[n]$, the output will be

$$|y[n]| = \left| \sum_{k=-\infty}^{+\infty} h[k]x[n-k] \right| \leq \sum_{k=-\infty}^{+\infty} |h[k]||x[n-k]| \leq B \sum_{k=-\infty}^{+\infty} |h[k]|$$

- $y[n]$ is bounded in magnitude, and hence is stable if $\sum_{k=-\infty}^{+\infty} |h[k]| < \infty$

- For continuous-time LTI systems the stability condition is $\int_{-\infty}^{+\infty} |h(\tau)| d\tau < \infty$

Frequency response of a LTI system

Discrete-time LTI system

- A discrete-time signal $x[n]$ can be decomposed into a sum of scaled and phase shifted complex exponentials.
- Consider a simple complex exponential signal $x[n] = e^{j\omega n}$, and $h[n]$ be the impulse response.
- The output of the LTI system:

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]\,h[n-k] \quad = \sum_{k=-\infty}^{+\infty} h[k]\,x[n-k]$$

$$= \sum_{k=-\infty}^{+\infty} h[k]e^{j\omega(n-k)} \quad = e^{j\omega n}\sum_{k=-\infty}^{+\infty} h[k]e^{j\omega(-k)}$$

$$= \boxed{H(e^{j\omega})}e^{j\omega n}$$

Frequency response

$$\boxed{H(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} h[k]e^{-j\omega k} = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})}}$$

Frequency response of a LTI system

Continuous-time LTI system

- Consider an exponential signal $x(t) = e^{j[\omega t+\theta]}$

$$y(t) = \int_{-\infty}^{+\infty} x(t-\tau)\, h(\tau) d\tau$$

$$= \int_{-\infty}^{+\infty} e^{j(\omega(t-\tau)+\theta)}\, h(\tau) d\tau$$

$$= e^{j(\omega t+\theta)} \int_{-\infty}^{+\infty} e^{-j\omega\tau}\, h(\tau) d\tau$$

$$= e^{j(\omega t+\theta)} \boxed{H(\omega)}$$

If the input of a LTI system is a sinusoidal signal, the output will also be a sinusoidal signal with the same frequency, but possibly shifted and multiplied by a constant.

Frequency response $H(\omega) = \int_{-\infty}^{+\infty} e^{-j\omega\tau}\, h(\tau) d\tau$

Frequency response of a LTI system

- **Response to an arbitrary signal**
  - If the input signal is represented as $x(t) = \sum_{k=1}^{K} c_k e^{j(\omega_k t + \theta_k)}$
  - The output signal:

$$y(t) = \mathcal{H}\big(x(t)\big) = \mathcal{H}\left(\sum_{k=1}^{K} c_k e^{j(\omega_k t + \theta_k)}\right)$$

$$= \sum_{k=1}^{K} \mathcal{H}\big(c_k e^{j(\omega_k t + \theta_k)}\big) \qquad = \sum_{k=1}^{K} c_k \mathcal{H}\big(e^{j(\omega_k t + \theta_k)}\big)$$

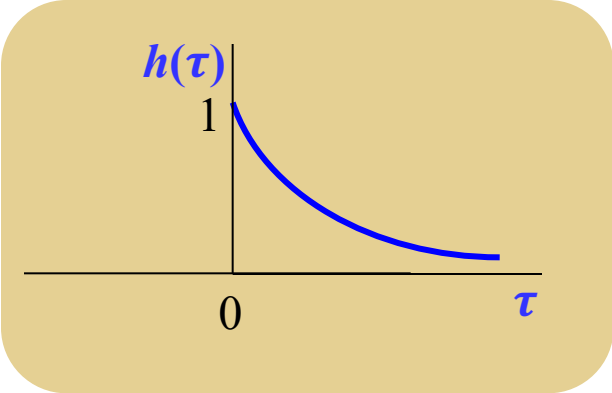
$$= \sum_{k=1}^{K} c_k \mathcal{H}(\omega_k) e^{j(\omega_k t + \theta_k)} \qquad = \sum_{k=1}^{K} c'_k e^{j(\omega_k t + \theta_k)}$$

$$\boxed{c'_k = c_k \mathcal{H}(\omega_k)}$$

Frequency response

Frequency response of a LTI system: Example

- **Find the frequency response of a linear time-invariant system with impulse response $h(t) = e^{-at}u(t)$.**



- **Answer:**

$$h(t) = e^{-at}u(t) \qquad H(\omega) = \int_{-\infty}^{+\infty} e^{-j\omega\tau}\, h(\tau)d\tau$$

$$H(\omega) = \int_{-\infty}^{+\infty} e^{-j\omega\tau}\, e^{-a\tau}u(\tau)d\tau$$

$$= \int_{0}^{+\infty} e^{-j\omega\tau}\, e^{-a\tau}d\tau$$

$$= \left.\frac{e^{-(a+j\omega)\tau}}{-(a+j\omega)}\right|_{0}^{+\infty} = \left.\frac{e^{-a\tau+j\omega\tau}}{-(a+j\omega)}\right|_{0}^{+\infty} = \frac{1}{a+j\omega}$$

System analysis: Time-based versus Frequency-based

Time-based approach

- Evaluate the impulse based response $h(t)$ by conducting an experiment or using system equation
- Find the output for an arbitrary input signal $x(t)$ by convolving it with the impulse response

$$y(t) = x(t) * h(t)$$

Frequency-based approach

- Represent the input signal $x(t)$ as a linear combination of sinusoidal signals

$$x(t) = \sum_{k=1}^{K} c_k e^{j(\omega_k t + \theta_k)}$$

- Compute the frequency response $\mathcal{H}(\omega)$
- The output is

$$y(t) = \sum_{k=1}^{K} c'_k e^{j(\omega_k t + \theta_k)}$$

$$c'_k = c_k \mathcal{H}(\omega_k)$$

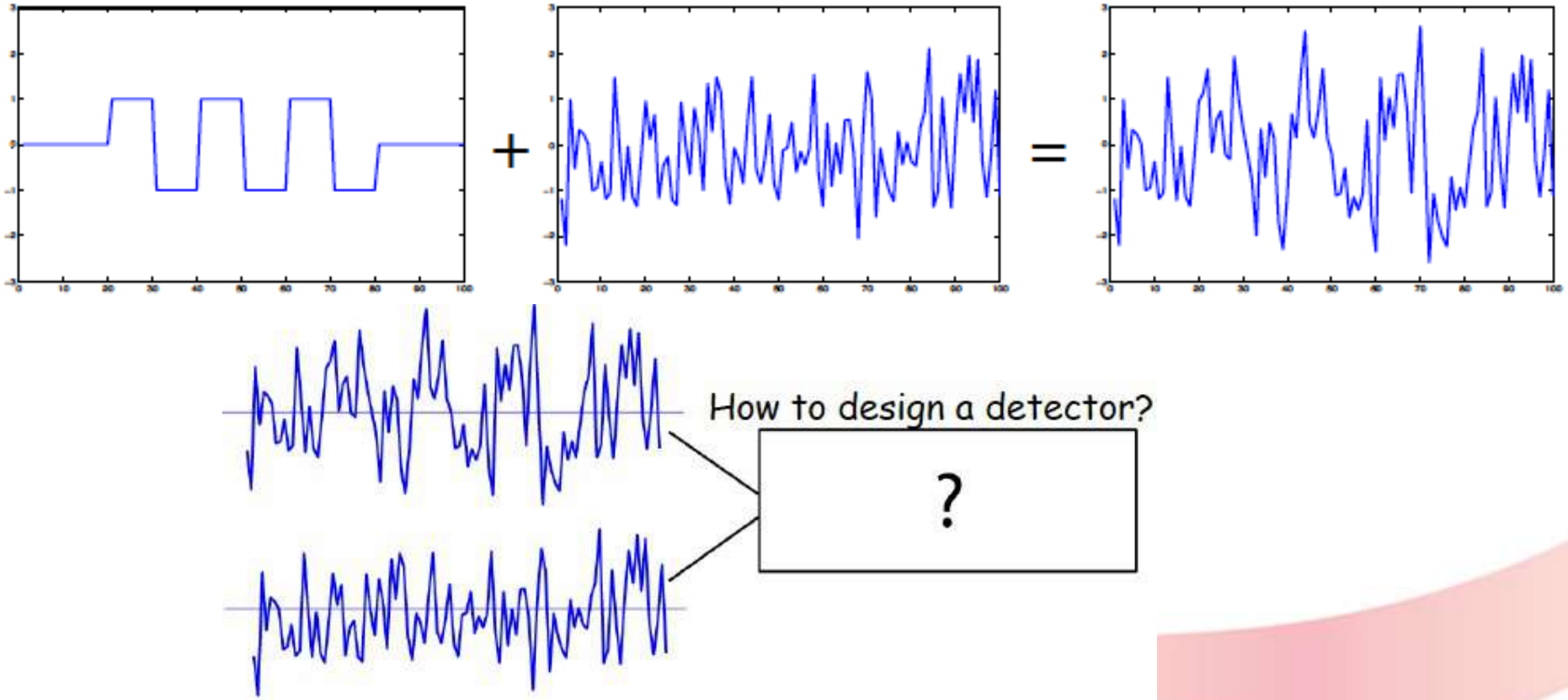
Modeling of Linear time-invariant systems

- **Modeling of LTI systems**
  - Based on the desired system properties, by determining its impulse response
  - Design a system that performs convolution of the input signal with the impulse response of the system

- **Examples:**
  - Signal detection using matched filter
  - Edge detection system in image processing
  - System to address multipath problems in communication systems

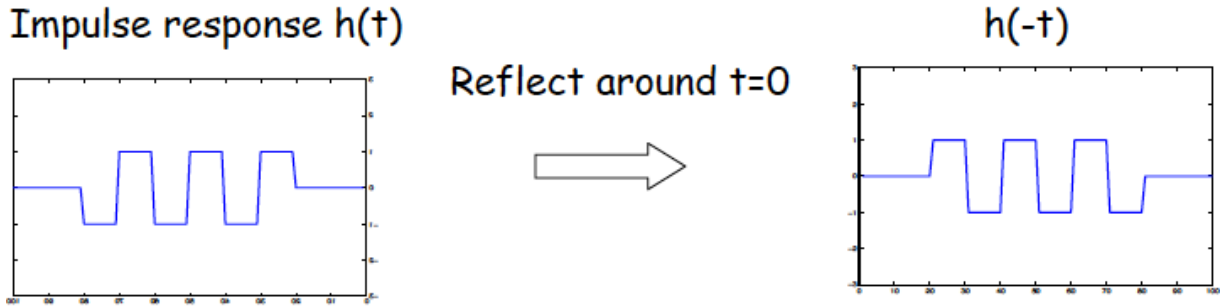
LTI system modeling: Examples

- **Signal detection using matched filter**
  - Goal: Detect signal in noise

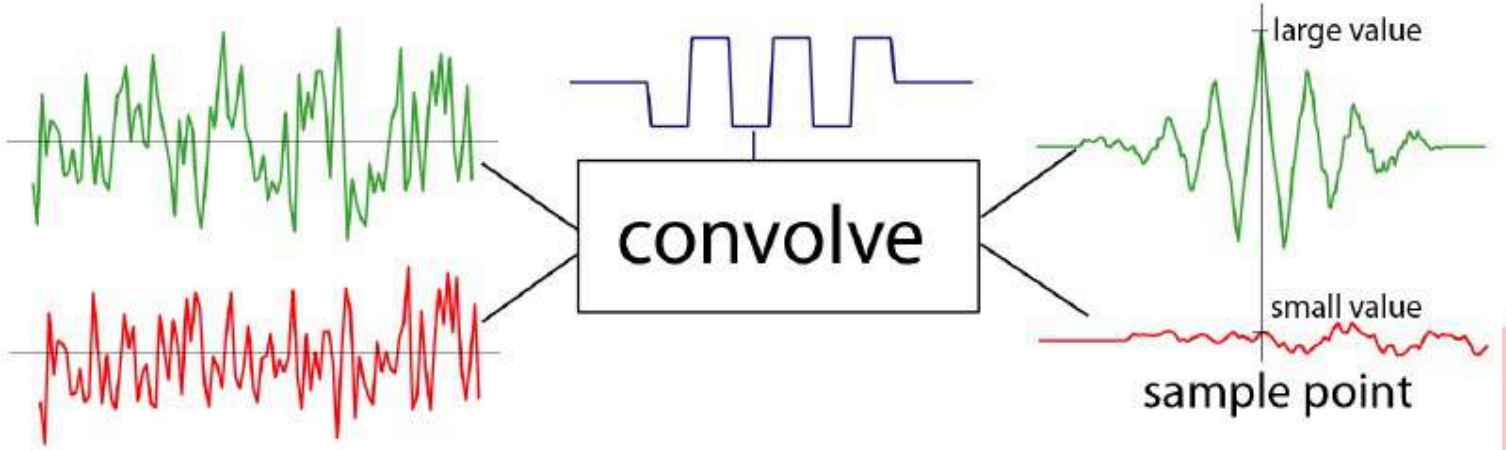


LTI system modeling: Examples

- **Signal detection using matched filter**
  - Design a matched filter corresponding to the signal

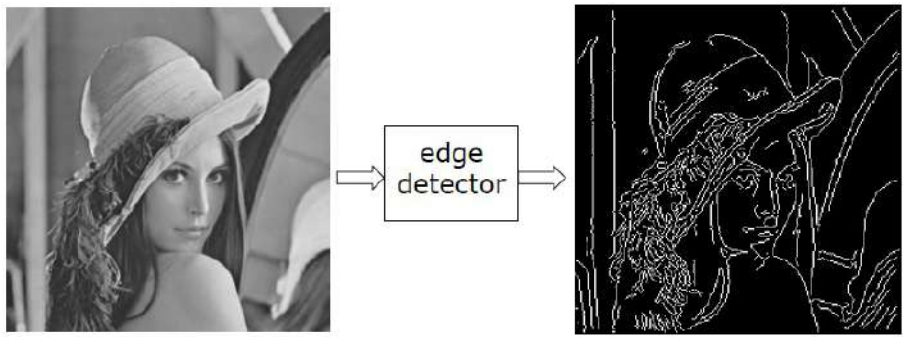


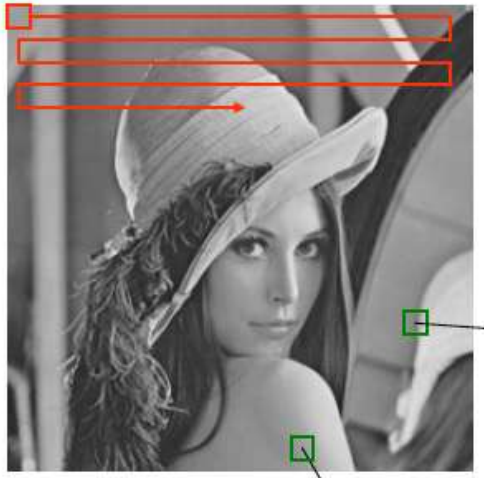
  - Perform a convolution of the matched filter with the received signal



LTI system modeling: Examples

- **Edge detection in image processing**
  - Goal: Detect image edges





impulse response h(m,n)

$$\begin{matrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{matrix}$$

h(-m,-n)

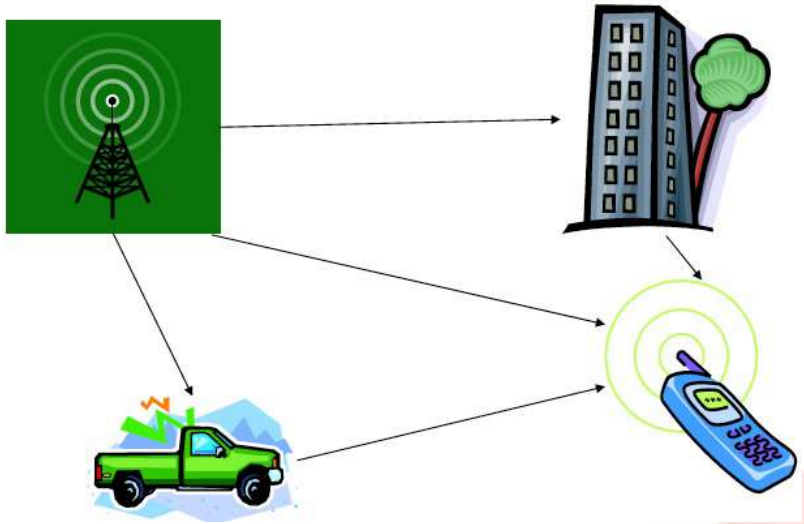
$$\begin{matrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{matrix}$$

$$\sum \left(\begin{matrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{matrix} \times \begin{matrix} 0.2 & 0.5 & 0.8 \\ 0.2 & 0.5 & 0.8 \\ 0.2 & 0.5 & 0.8 \end{matrix} \right) = 1.8$$

$$\sum \left(\begin{matrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{matrix} \times \begin{matrix} 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \end{matrix} \right) = 0$$

LTI system modeling: Examples

- **A wireless communication system in a multipath environment**
  - Designing a mobile receiver for wireless communication system
  - Communication signals in a wireless environment are reflected, scattered and diffracted due to the static objects in the medium
  - The received signal is a linear combination of direct and reflected (and thus delayed) signals
  - The medium (wireless channel) is a linear time-invariant system



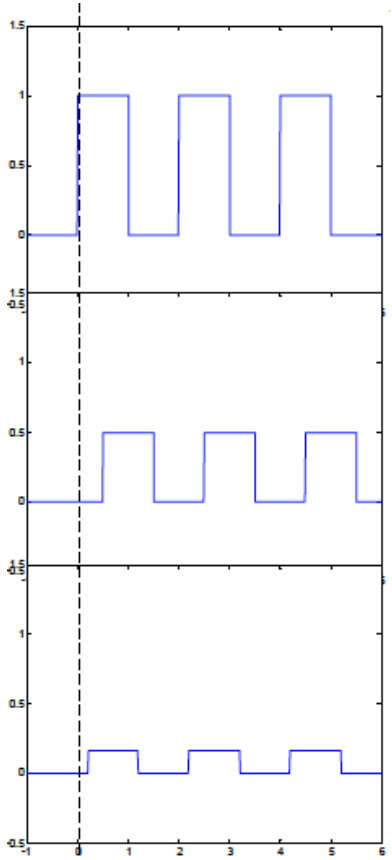
LTI system modeling: Examples

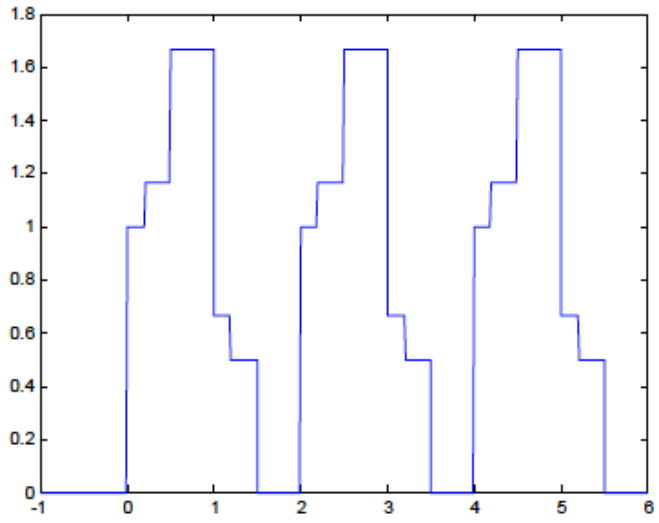
- **A wireless communication system in a multipath environment**
  - Results in signal superposition

Direct signal

Reflected signal 1
Building attenuation = 0.5
Delay = 0.5 sec

Reflected signal 2
Car attenuation = 0.16
Delay = 0.2 sec





Use this signal shape to construct a matched filter for the signal detection

Summary: Systems

- **What is a system?**
  - Refers to any physical device that produces an output signal in response to an input signal
- **Representation of systems**
  - Two main system representations: (i) via input-output functional relationships and (ii) via system equation that involves current and past values of the input
- **Classification of systems**
  - Continuous and discrete-time systems
    - Classification based on the time variable of input signal: continuous or discrete time
  - Memoryless systems and systems with memory
    - A system is memoryless if its output at a given time is dependent only on the input at that same time
    - Output of a system with memory depends on its past and/or future inputs either
  - Causal and noncausal systems
    - Output of causal systems depends only on present and past inputs, noncausal system's output depends on the future inputs

Summary: Systems

- **Classification of systems**
  - Linear and non-linear systems
    - A system is called linear if adding or scaling inputs to the system results in adding or scaling the corresponding outputs
  - Time variant and invariant systems
    - A system is time-invariant if delaying the input by arbitrary t_0 produces the output delayed by t_0
- **Linear time-invariant systems**
  - System is called linear time-invariant (LTI) if it is both linear and time-invariant
  - Majority of naturally occurring and man-made systems can be modeled as LTI systems
  - General form of a LTI system
    - Output of a LTI system is a linear combination of delayed and/or advanced version of the input
    - With the properties of superposition and time-invariance, one can develop a complete characterization of any LTI systems in terms of its response to a unit impulse

Summary: Systems

- **Analysis of Linear time-invariant systems**
  - Analysis of discrete-time systems
    - Discrete-time input signal for a given system can be represented as a linear combination of Kronecker delta functions
  - Computing the output of a discrete-time LTI system $y[n]$
  - Impulse response and convolution for a discrete-time system
    - Impulse response $h[n]$ of a discrete-time LTI system is the response of the system to the Kronecker delta function taken at zero initial conditions
    - Given the input signal $x[n]$, the output $y[n]$ of a discrete-time LTI system can be computed by convolving $x[n]$ with $h[n]$
  - Analysis of continuous-time systems
    - Any continuous-time signal can be represented as a finite weighted sum of shifted delta functions
  - Computing the output of continuous-time LTI system
  - Impulse response and convolution for a continuous-time system

Summary: Systems

- **Properties of LTI systems**
  - Commutative property of LTI systems
    - Convolution in both continuous and discrete time is a commutative operation
  - Distributive property of LTI systems
    - Using this, convolution can be exploited to break a complicated convolution into several simpler ones
  - Associative property
    - For LTI systems, the change of order of the cascaded systems will not affect the response
  - Invertibility of LTI systems
    - A system S is invertible if and only if there exists an inverse system S^{-1}such that $S^{-1}S$ is an identity system
  - Stability of LTI systems
    - A system is stable if every bounded input produces a bounded output

Summary: Systems

- **Frequency response of a LTI system**
  - Discrete-time LTI system
    - The signal $x[n]$ can be decomposed into a sum of scaled and phase shifted complex exponentials
  - Continuous-time LTI system
    - If the input of a LTI system is a sinusoidal signal, the output will also be a sinusoidal signal with the same frequency, but possibly shifted and multiplied by a constant
  - Comparison between time domain and frequency domain system analysis
- **Modeling of LTI systems**
  - Based on the desired system properties, and by determining its impulse response
  - Design a system that performs convolution of the input signal with the impulse response
  - Examples of LTI systems